



Web application frameworks

All you need to know



Jargon fills our everyday lives on the Internet. The days when HTML and JavaScript used to be ‘cool’ technologies are long gone. They are still very much an underlying part of the Net, but Web developers and designers create websites in a more powerful and sophisticated manner today—so much so that the term ‘website’ does not do justice to the systems that deliver content to our Web browsers. The ‘websites’ we visit aren’t just static information, but live and dynamic forms of information that interact with us. Hence, in the world of frameworks, they are called Web applications. I will explain the key concepts of this domain, so that by the time you are done reading this article, you will be able to hold a meaningful conversation on Web Application Frameworks.

Let me begin with the concept of the framework. If you look at the Wikipedia definition, it’s filled with even more unfamiliar terms. Hence, I will break it down into simple language for you. Imagine a cube made of only wires (like the structure of graphite that we all studied at school). This is called a wire-frame model of the cube. We can stick paper sheets onto the frames, wrap threads on the wires, paint them, and do lots of other modifications. We can do these because the model, or the cube (the ‘wire-frame’), allows us to. If I were to port this metaphor to the software world, the model of the cube (a bunch of wires) would become a collection of libraries. Libraries are pieces of code offering a specific set of functionalities. Now, just like the cube example, the way to use these libraries is through software functions (or ‘methods’ for all ye Java folk!). These functions are called the API (Application Programming Interface) of the particular framework. Thus, a software framework is a collection of libraries that can be combined with custom (i.e., user) code through an API.

Extending this concept to Web applications, there are frameworks available that are called (you guessed it) Web Application Frameworks. The libraries include functionalities like data access (to databases), session management (for multi-user capabilities), etc.

What do Web application frameworks do?

These have a wide array of features, some of which I will describe here:

- **URL mapping:** Frameworks, owing to their power, tend to have complex engines. In order for the engine to understand the desired operation, the URLs used in frameworks can get ugly (for the human user, as well as for search indexers, crawlers, etc). To simplify the URL into a more legible form, the URL mapping feature comes to the rescue. For example, you may have seen, on LFY’s earlier website, a URL like `www.lfymag.com/index.asp?id=13`. Instead of this, what if it was mapped to `www.lfymag.com/home`? That would not only be simpler to understand for the user, but for a search crawler too. In short, URL mapping is URL simplification.
- **Web templates:** A template is nothing but a predefined structure into which we can plug data or information as required. In the context of the framework, this is code that generates the remaining content on-the-fly, as required, on top of a predefined static set of pages. This way, the actual number of hard-coded pages drops drastically, and the performance of the application goes up significantly. Moreover, the data is handled more